Supplement: Dictionary-Based Slang Normalization for Indonesian Sentiment Classification

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These tables accompany the main manuscript and use its frozen held-out predictions. O/N denotes training on original inputs and evaluating normalized inputs; the first symbol always denotes the training condition. Scores for the three IndoBERT seeds use the same held-out reviews and are not independent test samples. No additional model was trained or selected for these tables.

Artifact Availability and Reproduction

The public `review_snapshot_v5`, in the repository linked in the main paper's Acknowledgement, contains this manuscript revision, code, configurations, aggregate results, a reviewer guide, and SHA-256 file manifests. It excludes review-level inputs or predictions, dictionary mappings, weights, private payloads, and credentials. Public access does not grant a project code license. `REPRODUCIBILITY.md` distinguishes synthetic checks and PDF/figure builds from full training or exact replay, which require additional resources.

PRDECT-ID's source states CC BY 4.0; the IndoBERT card states MIT. The pinned IndoNLU license file says Apache 2.0 but its README displays an MIT badge; SmSA terms remain unresolved. No explicit Kamus Alay license was identified.

Pinned Resources and Protocol

PRDECT-ID: DOI `10.17632/574v66hf2v.1`.
SmSA/IndoNLU revision:
`ce728f6926a36174b9923dfe49d6a6839b6e9bb7`
Kamus Alay revision:
`c710fa9f3f8d8727c24f3209f4f6a4b6ecaf6d8b`
IndoBERT base phase 1 revision:
`c2cd0b51ddce6580eb35263b39b0a1e5fb0a39e2`
Joint data/preprocessing fingerprint (concatenate lines):
`56a11ac3a8b5a70634a2b10af9f0f6c867b`
`50cc298685d60001979d3eb86884a`

Retrieval URLs and file hashes are in `data/manifest.json`; the 20-file audit lock is in `data/protocol_lock.json`. It identifies the data/preprocessing audit, not the later model selection.

Representation-Change Diagnostics

A post hoc vector-survival diagnostic found that, among text-changed test reviews, word TF-IDF vectors changed for 488/489 under the original-fitted PRDECT-ID vocabulary and 489/489 under the normalized-fitted vocabulary; vectors changed for 95/97 under either fixed SmSA word vectorizer. Under either fixed character vectorizer, they changed for 489/489 and 97/97, respectively. These counts show any vector change, not the fate of each replacement.

TABLE S1
IndoBERT Matched-Pipeline Results by Seed

Dataset	Seed	O/O	N/N	N/N − O/O
PRDECT-ID	13	0.9848	0.9734	−0.0114
PRDECT-ID	37	0.9823	0.9886	+0.0063
PRDECT-ID	73	0.9861	0.9810	−0.0051
SmSA	13	0.8905	0.9058	+0.0153
SmSA	37	0.8858	0.9008	+0.0150
SmSA	73	0.8726	0.9029	+0.0303

Differences are paired by seed; sample standard deviations are 0.0090 for PRDECT-ID and 0.0088 for SmSA. The same test reviews recur across seeds; seed variability is distinct from conditional test-sample intervals.

TABLE S2
Matched-Pipeline Class F1 and Support

Dataset	Model	Class	n	O/O	N/N
PRDECT-ID	Word SVM	Negative	412	0.9593	0.9666
PRDECT-ID	Word SVM	Positive	381	0.9547	0.9626
PRDECT-ID	Character SVM	Negative	412	0.9618	0.9631
PRDECT-ID	Character SVM	Positive	381	0.9572	0.9585
PRDECT-ID	IndoBERT	Negative	412	0.9852	0.9820
PRDECT-ID	IndoBERT	Positive	381	0.9836	0.9801
SmSA	Word SVM	Negative	204	0.8426	0.8337
SmSA	Word SVM	Neutral	88	0.5263	0.5674
SmSA	Word SVM	Positive	208	0.8363	0.8232
SmSA	Character SVM	Negative	204	0.8145	0.8238
SmSA	Character SVM	Neutral	88	0.5507	0.5612
SmSA	Character SVM	Positive	208	0.7990	0.8103
SmSA	IndoBERT	Negative	204	0.9372	0.9495
SmSA	IndoBERT	Neutral	88	0.7769	0.8248
SmSA	IndoBERT	Positive	208	0.9348	0.9352

Support counts distinct test reviews. IndoBERT F1 is the mean of seed-specific F1 scores, not a pooled prediction score.

TABLE S3
FIXED-MODEL MACRO-F1 CHANGES AND PREDICTION TRANSITIONS

Dataset	Model	Direction	Macro-F1 change	95% CI	Corrected	Introduced
PRDECT-ID	Word SVM	$O \to N$	+0.0013	[−0.00883, +0.01268]	10 / 10	9 / 9
PRDECT-ID	Word SVM	$N \to O$	−0.0277	[−0.04285, −0.01266]	7 / 7	29 / 29
PRDECT-ID	Character SVM	$O \to N$	+0.0051	[−0.00003, +0.01145]	5 / 5	1 / 1
PRDECT-ID	Character SVM	$N \to O$	−0.0113	[−0.02392, +0.00017]	8 / 8	17 / 17
PRDECT-ID	IndoBERT	$O \to N$	−0.0008	[−0.00339, +0.00169]	0.67 / 1	1.33 / 3
PRDECT-ID	IndoBERT	$N \to O$	+0.00001	[−0.00294, +0.00294]	1.67 / 5	1.67 / 4
SmSA	Word SVM	$O \to N$	+0.0016	[+0.00000, +0.00497]	1 / 1	0 / 0
SmSA	Word SVM	$N \to O$	−0.0056	[−0.01681, +0.00171]	1 / 1	3 / 3
SmSA	Character SVM	$O \to N$	+0.0030	[+0.00000, +0.00762]	2 / 2	0 / 0
SmSA	Character SVM	$N \to O$	+0.0025	[−0.00453, +0.01275]	1 / 1	1 / 1
SmSA	IndoBERT	$O \to N$	+0.0006	[−0.00151, +0.00335]	0.33 / 1	0.33 / 1
SmSA	IndoBERT	$N \to O$	+0.0023	[−0.00007, +0.00537]	1.33 / 4	0 / 0

Transition counts show mean / unique; macro-F1 changes average paired seed differences for IndoBERT. Corrected and introduced mean incorrect-to-correct and correct-to-incorrect. IndoBERT means average across seeds; unique counts are unions of review IDs within each category and can overlap across categories. $O \to N$ fixes the original-trained model; $N \to O$ fixes the normalized-trained model. Their differences are $O/N - O/O$ and $N/O - N/N$, respectively. The near-zero positive PRDECT-ID IndoBERT difference uses five decimals; other changes use four; interval bounds use five. Intervals are post hoc, exploratory, nominal 95% percentile intervals from the same 2,000 paired test-group draws (seed 42) as the primary evaluation. Each draw averages paired IndoBERT seed effects. They condition on fitted models, exclude training uncertainty, and have no simultaneous coverage; intervals containing zero do not establish equivalence.

TABLE S4
RETAINED CLASS SUPPORT BY SPLIT

Dataset	Split	Negative	Neutral	Positive
PRDECT-ID	Train	1,922	–	1,780
PRDECT-ID	Validation	412	–	381
PRDECT-ID	Test	412	–	381
SmSA	Train	3,407	1,136	6,376
SmSA	Validation	394	131	735
SmSA	Test	204	88	208

Counts are distinct reviews. PRDECT-ID has no neutral class; splits and labels were fixed before evaluation.

TABLE S5
VALIDATION-SELECTED SETTINGS AND FINAL INDOBERT EPOCHS

Dataset	Condition	SVM C	IndoBERT LR	Epochs
PRDECT-ID	O	1 / 1	2×10^{-5}	2 / 3 / 3
PRDECT-ID	N	1 / 1	2×10^{-5}	3 / 2 / 4
SmSA	O	1 / 1	2×10^{-5}	3 / 5 / 2
SmSA	N	10 / 1	3×10^{-5}	2 / 4 / 5

SVM C values are Word / Character. Epochs are best validation epochs for final seeds 13 / 37 / 73. Learning rates and C were selected on validation; no test-based retuning was performed.

TABLE S6
TUNING AND TRAINING SETTINGS

Component	Setting
Linear SVM	$C \in \{0.1, 1, 10\}$; unweighted classes; squared hinge loss with L_2 penalty
IndoBERT search	Learning rate $\in \{2, 3, 5\} \times 10^{-5}$; tuning seed 42
Optimization	AdamW; weight decay 0.01; effective batch size 16; gradient clipping at 1.0
Schedule	Five epochs; 10% linear warmup, followed by linear decay
Selection	Highest validation macro-F1; evaluation after each epoch for IndoBERT
Final runs	SVM: seed 42; IndoBERT: seeds 13, 37, 73